

Polynomial Regression Assignment Report

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1. Summary

Two polynomial regression models were trained for the assigned datasets. The final model for var1 used a degree-5 polynomial expansion with ElasticNet regularization, while the final model for var2 used a degree-10 polynomial expansion with ElasticNet regularization.

Problem	Degree	Final model	10-fold CV MSE	10-fold CV R2
var1: Steam turbine optimization	5	ElasticNet, alpha = 0.009103, l1_ratio = 1.0	0.297031 +/- 0.023987	0.971316 +/- 0.004937
var2: Thermal reservoir mapping	10	ElasticNet, alpha = 0.000655, l1_ratio = 0.1	0.230232 +/- 0.041747	0.994225 +/- 0.002097

2. ElasticNet Objective

ElasticNet was fitted on the polynomial feature matrix. The objective minimized was:

$$\text{MSE}(y, X_{\text{poly}} \beta) + \alpha * [\text{l1_ratio} * ||\beta||_1 + (1 - \text{l1_ratio}) * ||\beta||_2^2 / 2]$$

Here, alpha controls the overall regularization strength and l1_ratio controls the balance between L1 sparsity and L2 shrinkage.

3. Data Exploration and Preprocessing

The var1 dataset has six input variables and var2 has three input variables. Inputs were already normalized to approximately [-1, 1], but scaling was still applied after polynomial expansion because high-degree terms can have different variances.

The pipeline generated polynomial features using PolynomialFeatures(include_bias=False), standardized them with StandardScaler, and fitted a regularized linear model. Keeping these steps inside a pipeline ensured fold-wise scaling during cross-validation and avoided leakage.

Problem	Target mean	Target std	Target min	Target max	Largest linear correlations with y
var1	0.8736	3.2708	-10.7271	11.3392	x5: -0.255, x6: -0.227
var2	2.0807	6.6172	-29.9015	32.3655	x2: 0.457, x3: 0.190

Exploratory analysis showed weak individual feature-target correlations in var1 and a wider target spread in var2. These observations supported the use of polynomial feature expansion to capture nonlinear effects and interactions.

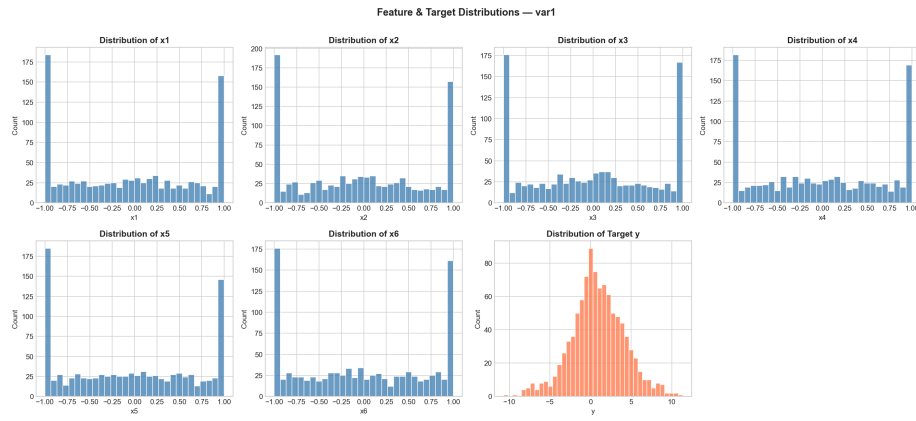


Figure 1a: var1 feature and target distributions.

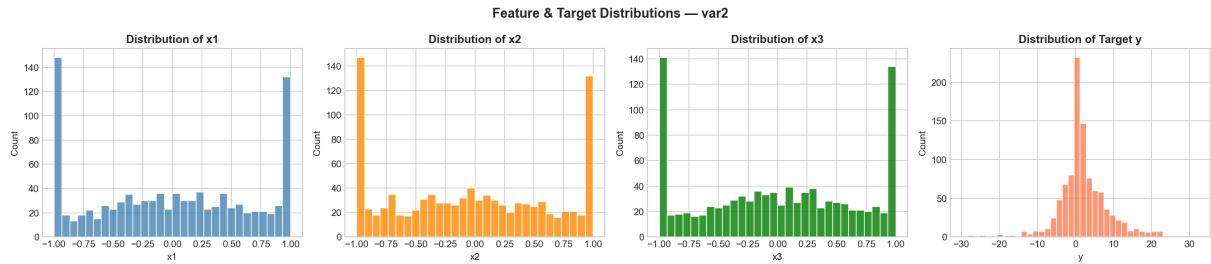


Figure 1b: var2 feature and target distributions.

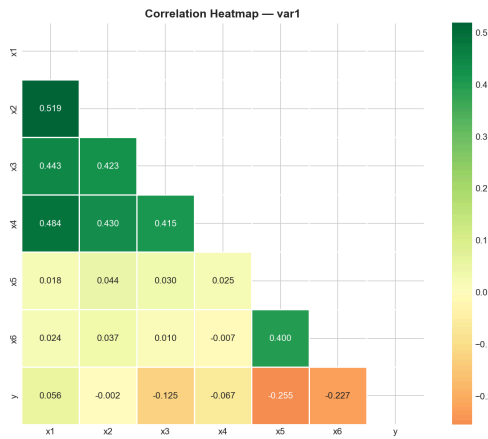


Figure 2a: var1 correlation heatmap.

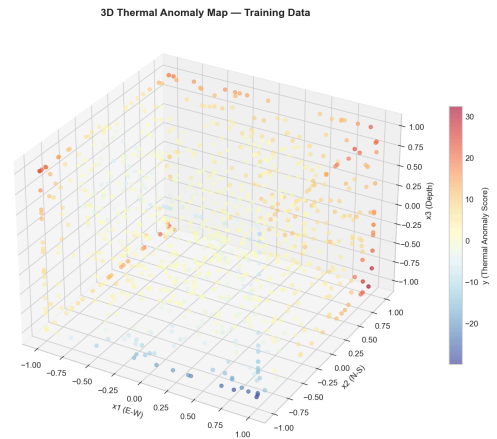


Figure 2b: var2 3D training scatter colored by target value.

4. Model Selection Approach

The two main modeling decisions were polynomial degree and regularization strength. Ridge regression with $\alpha=1.0$ was used as a stable proxy during degree selection because polynomial expansion creates correlated features. After choosing the degree, ElasticNet was tuned over α and $l1_ratio$.

ElasticNet was useful because L2 regularization stabilizes coefficients, while L1 regularization can remove unnecessary polynomial terms and reduce overfitting.

5. Polynomial Degree Selection

var1

For var1, degrees 1 to 10 were evaluated. Validation error decreased up to degree 5 and then increased for higher degrees. This indicates that degrees above 5 started overfitting the training data. Therefore, degree 5 was selected.

var1 — Degree Selection via 10-Fold CV

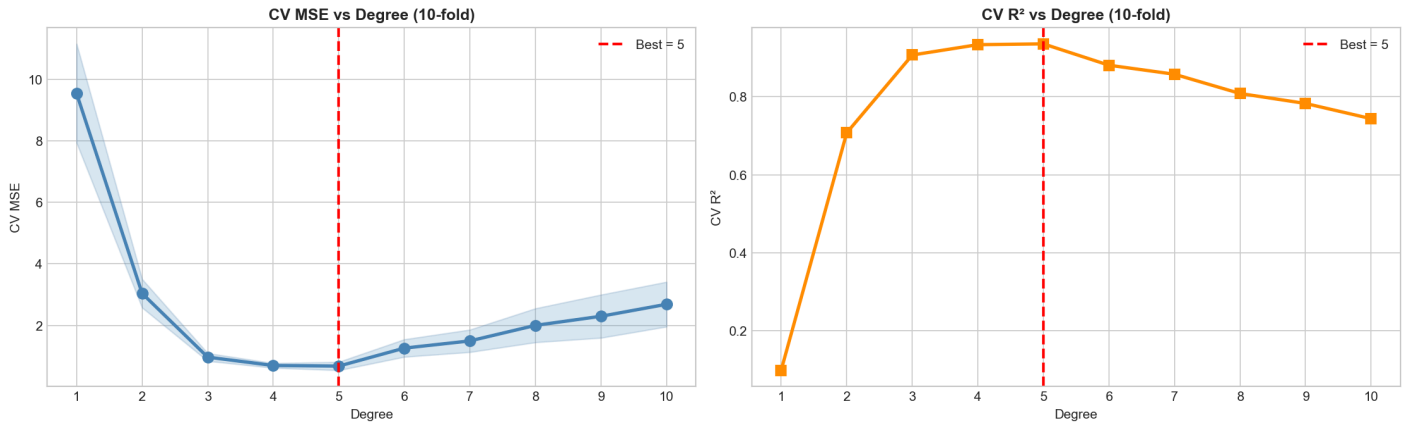


Figure 3: Degree selection for var1 using 10-fold cross-validation.

var2

For var2, degrees 1 to 20 were evaluated. The validation score improved until degree 10, after which the improvement stopped and the error slightly increased. Degree 10 was selected because it gave the best validation performance while avoiding unnecessary higher-degree complexity.

var2 — Degree Selection via 10-Fold CV

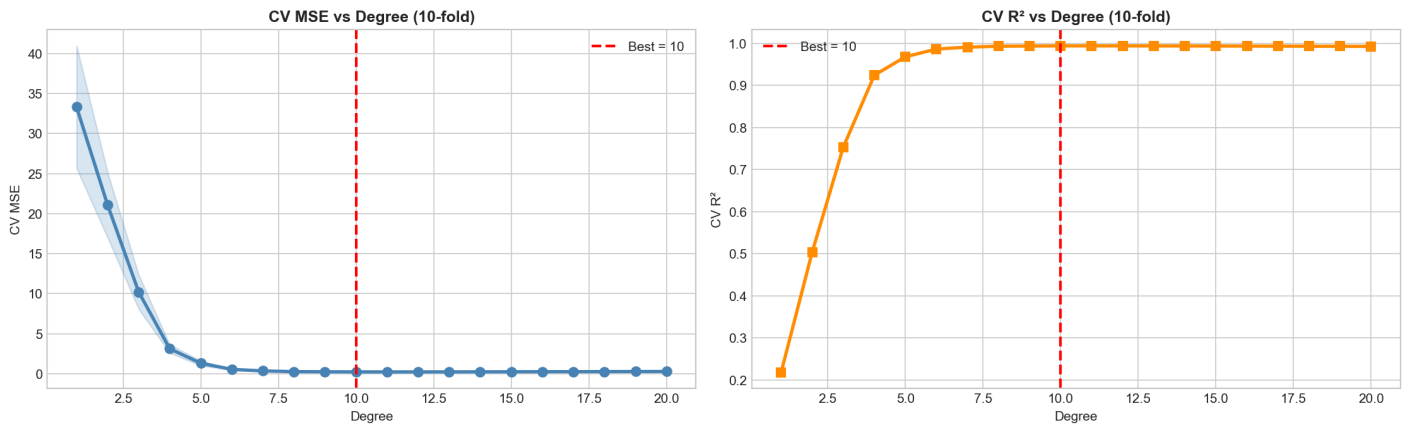


Figure 4: Degree selection for var2 using 10-fold cross-validation.

6. Final Model Details

var1 quantity	Value
Selected polynomial degree	5
Number of polynomial features	461
Best alpha	0.009103
Best l1_ratio	1.0
Non-zero coefficients	126 / 461
Training MSE	0.216335
Training R2	0.979757
10-fold CV MSE	0.297031 +/- 0.023987
10-fold CV R2	0.971316 +/- 0.004937

var2 quantity	Value
Selected polynomial degree	10
Number of polynomial features	285
Best alpha	0.000655
Best l1_ratio	0.1
Non-zero coefficients	253 / 285
Training MSE	0.162370
Training R2	0.996288
10-fold CV MSE	0.230232 +/- 0.041747
10-fold CV R2	0.994225 +/- 0.002097

For var1, `l1_ratio=1.0` made the model Lasso-like and removed many weak degree-5 terms. For var2, `l1_ratio=0.1` kept the model mostly Ridge-like, suitable for a response distributed across many degree-10 terms.

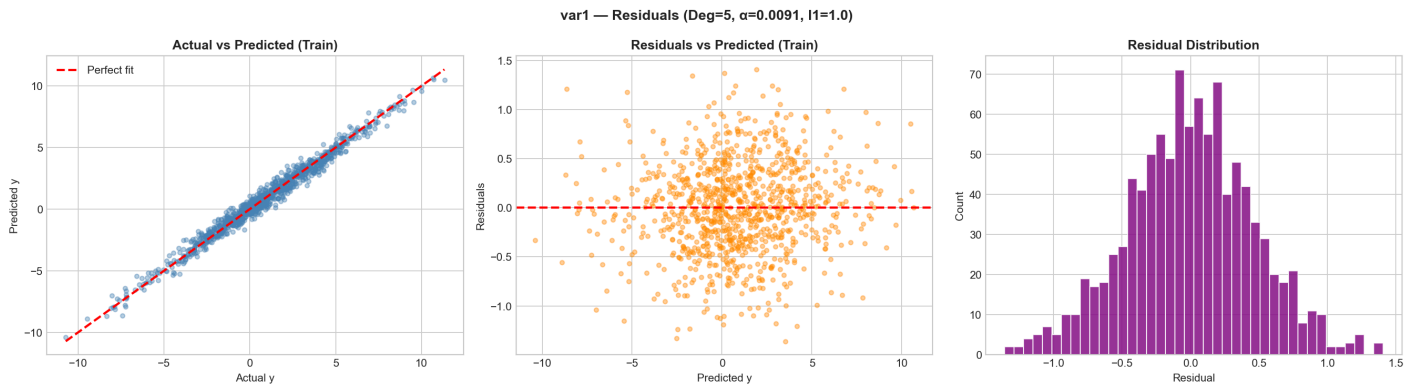


Figure 5a: var1 training residual diagnostics.

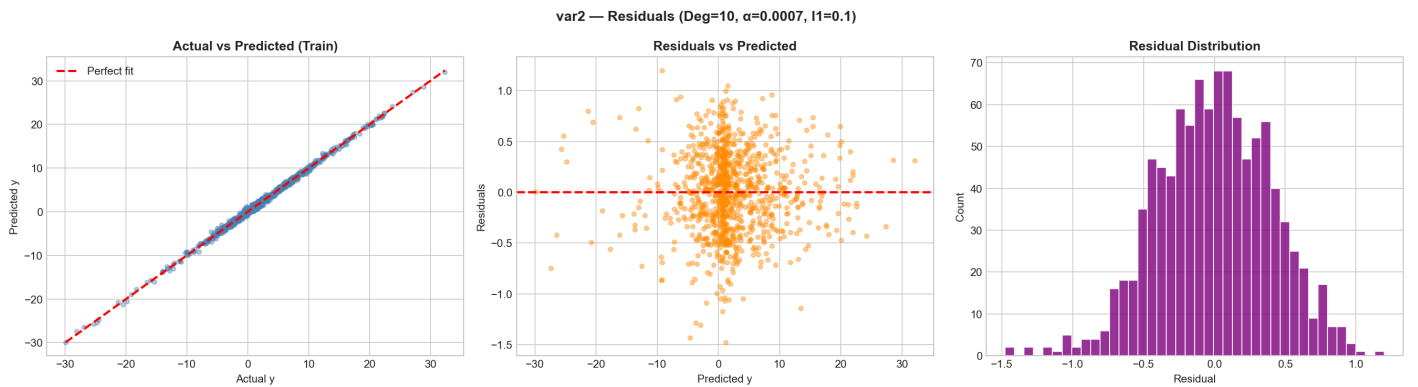


Figure 5b: var2 training residual diagnostics.

7. Prediction and Regularization Diagnostics

Prediction distributions were compared against training target distributions as a sanity check that the final polynomial models produced realistic ranges.

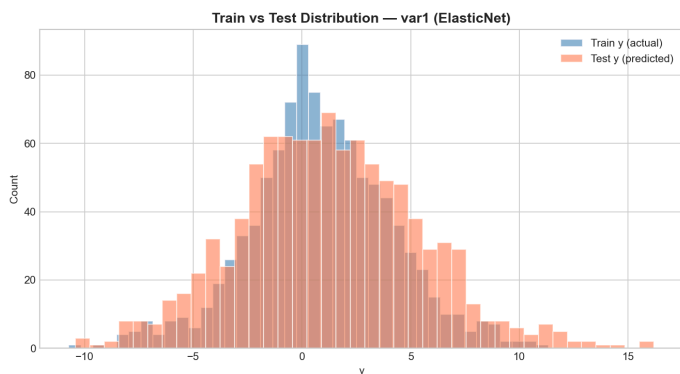


Figure 6a: var1 training target vs test prediction distribution.

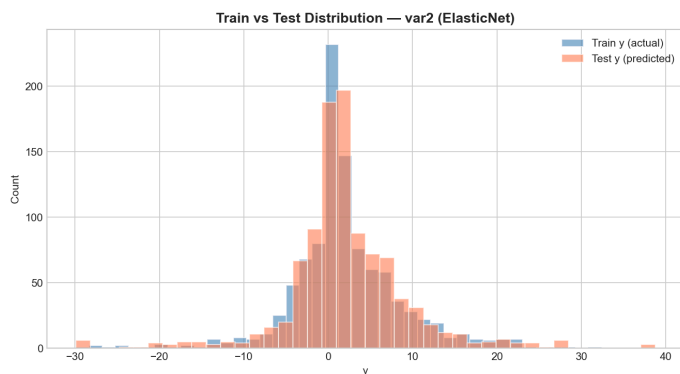


Figure 6b: var2 training target vs test prediction distribution.

The coefficient sparsity plots further explain the different regularization behavior. var1 is much sparser because the best model uses a Lasso-like penalty, while var2 keeps most polynomial coefficients active because its fitted response is more distributed across the degree-10 feature space.

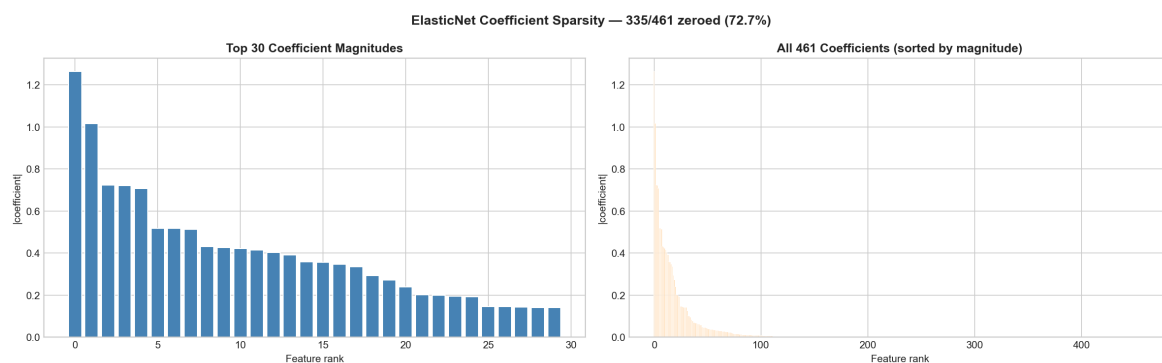


Figure 7a: var1 coefficient sparsity.

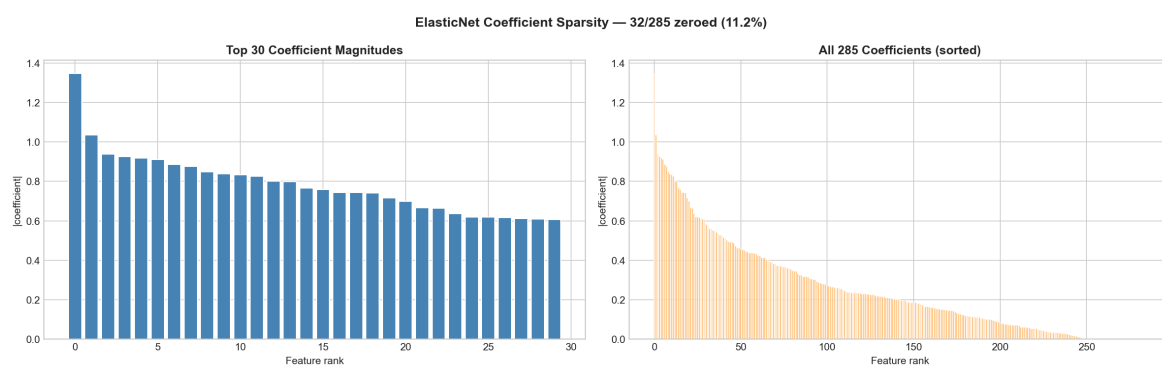


Figure 7b: var2 coefficient sparsity.

8. Conclusion

The final models are polynomial regression pipelines with cross-validated degree selection and ElasticNet regularization. Degree 5 was selected for var1 because it gave the best validation performance before overfitting began. Degree 10 was selected for var2 because it captured the higher-degree spatial structure while maintaining the best cross-validation score.

The final 10-fold CV R2 scores were 0.971316 for var1 and 0.994225 for var2. The two prediction CSV files were generated from the final fitted pipelines for assignment submission.